

Special Functions

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1 Orthogonal Polynomials

There exist $a < x_1 < x_2 < \cdots < x_n < b$ and $\lambda_i > 0$ such that for every polynomial f of degree at most $2n - 1$ that

$$\int_a^b f(x)w(x)dx = \sum_{i=1}^n \lambda_i f(x_i).$$

Proof: Let x_i be the roots of f_n . Let $\lambda_j = \int_a^b l_j(x)w(x)dx$. Write $f(x) = p_n(x)g(x) + r(x)$, where $\deg r \leq n - 1$. Then

$$\begin{aligned} \int_a^b f(x)w(x)dx &= \int_a^b p_n(x)g(x)w(x)dx + \int_a^b r(x)w(x)dx \\ &= \langle p_n(x), g(x) \rangle + \int_a^b \sum_{i=1}^n r(x_i)l_i(x)w(x)dx = \sum_{i=1}^n r(x_i) \int_a^b l_i(x)w(x)dx \\ &= \sum_{i=1}^n \lambda_i r(x_i) = \sum_{i=1}^n \lambda_i f(x_i). \end{aligned}$$

$l_j^2 - l_j$ is degree $2n - 2$. Note $l_j(x_j) = 1$, so $l_j^2 - l_j$ has roots x_k for $k \in [n]$. Write $l_j^2 - l_j = p_n \cdot h$, where $\deg h = n - 2$. Then

$$\int_a^b (l_i(x)^2 - l_i(x))w(x)dx = \int_a^b p_n(x)h(x)w(x)dx = \langle p_n, h \rangle = 0,$$

so

$$\lambda_i = \int_a^b l_i(x)w(x)dx = \int_a^b l_i^2(x)w(x)dx > 0.$$

Define the function $p_n(x) = \frac{1}{K_n w(x)} \left(\frac{d}{dx} \right)^n [w(x)(1-x^2)^n]$. Then $\int_{-1}^1 p_n(x)x^k w(x)dx = 0$ for all $0 \leq k < n$.

Proof:

$$\begin{aligned}
& \int_{-1}^1 \frac{1}{K_n w(x)} \left(\frac{d}{dx} \right)^n [w(x)(1-x^2)^n] x^k w(x) dx = \frac{1}{K_n} \int_{-1}^1 x^k \left(\frac{d}{dx} \right)^n [w(x)(1-x^2)^n] dx \\
&= \frac{1}{K_n} \left(x^k \left(\frac{d}{dx} \right)^{n-1} [w(x)(1-x^2)^n] - \int kx^{k-1} \left(\frac{d}{dx} \right)^{n-1} [w(x)(1-x^2)^n] dx \right) \Big|_{-1}^1 \\
&= -\frac{k}{K_n} \int_{-1}^1 x^{k-1} \left(\frac{d}{dx} \right)^{n-1} [w(x)(1-x^2)^n] dx \\
&= -\frac{k}{K_n} \int_{-1}^1 x^{k-1} \left(\frac{d}{dx} \right)^{n-1} [w(x)(1-x^2)^{n-1} - w(x)(1-x^2)^{n-1} x^2] dx.
\end{aligned}$$

We compute

$$\begin{aligned}
P_n^{(\alpha, \beta)}(x) &= \frac{(-1)^n}{2^n \cdot n!} \cdot \frac{1}{(1-x)^\alpha (1+x)^\beta} \frac{d^n}{dx^n} (1-x)^{\alpha+n} (1+x)^{\beta+n} \\
&= \frac{(-1)^n}{2^n \cdot n!} \cdot \frac{1}{(1-x)^\alpha (1+x)^\beta} \sum_{k=0}^n \binom{n}{k} (-1)^k (\alpha+n)_k (\beta+n)_{n-k} (1-x)^{\alpha+n-k} (1+x)^{\beta+k} \\
&= \frac{(-1)^n}{2^n \cdot n!} \sum_{k=0}^n \binom{n}{k} (-1)^k (\alpha+n)_k (\beta+n)_{n-k} (1-x)^{n-k} (1+x)^k,
\end{aligned}$$

so

$$P_n^{(\alpha, \beta)}(1) = \frac{(-1)^n}{2^n \cdot n!} (-1)^n (\alpha+n)_n 2^n = \binom{\alpha+n}{n}.$$

For $\alpha, \beta = 0$, we have

$$P_n^{(0,0)}(x) = \frac{(-1)^n}{2^n \cdot n!} \frac{d^n}{dx^n} (1-x^2)^n,$$

the Legendre polynomials.

$f \in L^2(-1, 1)$ iff $\int_{-1}^1 f(x)^2 dx < \infty$. Suppose $f(x) = \sum_{n=0}^{\infty} c_n P_n(x)$ uniformly converges on $[-1, 1]$. Then

$$\int_{-1}^1 f(x) P_m(x) dx = \sum_{n=0}^{\infty} c_n \int_{-1}^1 P_n(x) P_m(x) dx = \frac{c_m}{1+m/2},$$

$$\text{so } c_n = \left(n + \frac{1}{2}\right) \int_{-1}^1 f(x) P_n(x) dx.$$

Choose a_N such that $m_N = \int_{-1}^1 [f(x) - \sum_{n=0}^N a_n P_n(x)]^2 dx$ is as small as possible. Then the minimal values are achieved for $c_n = a_n$ for all $n \in \mathbb{Z}_{\geq 0}$.